Instructor key: Paper 006

AuthorityEffect: None. The quiz has 24 questions across levels 0-7 and a 96-point rubric. Accept equivalent calculations when their source, encoding, and evidence limits agree. This key is separate from the study guide so learners can attempt the examination first.

Level 0

Q01

The sink assigns index/timestamp, normalizes supplied data, computes digests, and writes a header and entries. It did not calculate the native functional or decide the truth of the supplied reason. `NO_NATIVE_PORT` describes missing infrastructure under its caller's contract, not a demonstrated mathematical defect. Bind checker/source or binary, actual input/output bytes, and an independent verification of the relationship before making a result claim.

Q02

Logical append is the interface's increasing ordered history. The physical file is replaced, not immutable. Successful focused restarts demonstrate the exercised reconstruction checks. No lock, explicit fsync, or power-loss campaign establishes concurrent-writer safety or general durability.

Q03

The value is the literal string `"None"`; it is not JSON `null`. A different entry marker is rejected by creation/reload checks. The required value grants no operational authority and authenticates no producer. A copied marker is not a signature.

Level 1

Q04

`Fraction(2, 6)` reduces to `Fraction(1, 3)` and is stored as `"1/3"`. The text value is already the same string. The protected value cannot distinguish their original types. Retain independently authorized original bytes and an explicit type/encoding contract if exact reconstruction matters.

Q05

The sanitized result is `{"1": "text"}`: the later string key overwrites the earlier key after both become `"1"`. Hashing occurs after this loss. It is a normalization collision, not two different encoded payloads with a SHA-256 collision. Specify allowed key types or retain the original representation.

Q06

U+1F600 has UTF-16 units D83D DE00, whose first unit is less than FB33; therefore it sorts before U+FB33 in JCS. Python code-point order is the reverse. The baseline encoder also permits NaN/infinity and does not implement JCS number serialization. Either is an acceptable second distinction. Later float tagging and rejection use different source bytes; bind them separately.

Level 2

Q07

Change	Local reload	Independent `(2,h1)`	Independent original raw-file digest
Protected reason changed, old digest kept	Integrity rejection	No verified altered snapshot	Differs
Final entry removed, old header kept	Truncation rejection	No verified altered snapshot	Differs
Final entry removed, count/head rewritten	Can pass with `(1,h0)`	Differs	Differs
Only header schema/genesis changed	Can pass at the baseline	Can match	Differs

The anchor must be held outside the rewriting adversary's control, and the argument assumes an alternative protected encoding cannot replace the expected digest. No conclusion about original types, producer honesty, or raw identity follows from normalized equality alone. For the schema case, the baseline loader's omission is part of the answer; the receipt's own schema does not validate the file header.

Q08

No. A legitimate extension also changes its head. Verify all three entries and compare entry index 1's digest with `h1` under the retained count two. Under the hash/custody assumptions this checks the normalized retained prefix. It does not authenticate the producer, the clock, unbound header fields, raw bytes, or the semantic correctness of the additional event.

Q09

The object emits its cached entry count/head, so it has not observed the file change. Reopen and emit only after successful verification, or call `verify_integrity` successfully before emission. The described alteration raises `LedgerIntegrityError`, yielding no verified changed snapshot. Assume one writer and custody that prevents a competing modification during the check.

Level 3

Q10

The exception is not an acknowledged append and not a rollback certificate. The cached receipt counts three; the reopened persisted file still counts two. Discard or successfully reload the old object and reconcile the attempted operation before retrying. After a successful reload the old object also returns to persisted count two. This particular injected failure does not settle all storage failures or crash behaviors.

Q11

No transitive immutability follows. The changed mapping can be written with its stale stored digest during a later append; subsequent reload rejects the changed protected payload. Frozen fields and a tuple do not freeze nested dictionaries. Verified bytes are the evidence source; reference discipline and immutable retention are additional obligations.

Q12

`1000 * (100*101/2) = 5,050,000` accumulated entry bytes, versus `1000*100 = 100,000` final entry bytes. Headers are excluded. Appropriate additional limits include record size, total retained bytes, memory, writer coordination, temporary-path interference, regex runtime, and write-failure recovery. This is an idealized cost count, not empirical throughput.

Level 4

Q13

There are 12 non-placeholder entries. Fractions are 5/61 labelled active and 49/61 carrying placeholders. These counts establish zero demonstrated reproductions or enforcement actions by themselves; this means no such demonstration follows from this evidence, not a measured count of failures of every possible detector. The catalog has 14 literal failure classes.

Q14

The digest covers the specified identifying string, not a reproduction receipt. Raw-event identity would need defined event bytes and their binding; semantic defect identity needs an additional declared equivalence/checker. An independently retained digest permits comparison of its covered bytes. Repeating one record 1,000 times adds stored copies, not 1,000 independently checked defects or draws from a population.

Q15

Under the irreflexive relation, the loop fails the declared predicate. Under the loop-permitting signed grammar, the same display is valid. Bind carrier, grammar, predicate, checker, and route/action scope; retaining the first refusal does not license universal suppression of that phrase. These two classroom declarations suffice without invoking an undeclared graph filling.

Level 5

Q16

True plus a kill receipt reports a benign-panel match and proposes the clone/parent purge. False plus no record reports no match on that panel. False plus an invalid-pattern refusal rejects the malformed clone while sparing its parent under this check. The Boolean alone is inadequate. Returning a record does not mutate either frozen object or prove a repertoire or host-policy change. A consumer must apply and record the authorized action.

Q17

The API index is 1/6, below 1/5, so no downgrade record is synthesized at the default threshold. Conditional on a pattern match, the fraction is 1/3, which would exceed 1/5, but that is not this interface's denominator. Signal presence is not an independent validity label or an observed suppression action. Population false suppression requires independently valid eligible proposals, actual actions, and an appropriate sampling/evaluation plan.

Q18

Corpus mode reports one pattern-only trigger, one total trigger, and toxicity 1. It synthesizes a downgrade at default tau. Empty explicit triggers give counts zero/zero and index zero by convention; this is no-sample evidence. Neither value alone certifies deployment safety, and a returned diagnostic record alone does not change enforcement. Distinguish pattern screening, oracle labels, and applied actions.

Level 6

Q19

Idle advances from 4 to 12. New completed intervals are `floor(12/5)-floor(4/5) = 2-0 = 2`; the score is `(4/5)/4 = 1/5`. Activity is false because `12 < 10` is false. Tau controls score spacing; `half_life_runs` controls idle expiry in this implementation.

Q20

Batch endpoints are 0, 3, 7, 12. Completed interval increments are 0, 1, 1; the successive scores are 1, 1/2, 1/4. One 12-run call crosses two intervals and also yields 1/4. In general, `sum(floor(t_j/tau)-floor(t_(j-1)/tau)) = floor(t_final/tau)-floor(t_initial/tau)`. The intermediate terms cancel. Keep tau fixed, elapsed counts valid, initial score nonnegative, and no resets/stimulation/manual score updates between batches. Changing those conditions changes the claim.

Q21

The score is `1/4 + 1/5 = 9/20`; idle is zero; activity is true. The method returns the modified candidate according to its maintenance policy. It does not independently validate an admission, prove useful reasoning, or show that a host installed the result.

Level 7

Q22

An acceptable packet binds original authorized bytes/encoding, event and host decision, source/checker or binary version, input/output hashes, independently retained anchor and custody, deterministic reproduction and result, a valid near miss with its independent judgment, predicate, fixture bank, permitted action, route/domain, and receipt for the new evaluation. Identifier hashes and placeholder strings leave these bindings/results missing. To demonstrate enforcement, record the consuming host's applied action against the admitted rule version and relevant before/after state, not merely a proposed downgrade.

Q23

Group duplicates/base contents and related source cases before train/holdout allocation. Freeze the detector before held-out evaluation; do not reuse its training near misses as independent tests. Count independently valid eligible proposals in the safety denominator and actual wrongful suppressions in its numerator. Report group diversity and justified dependence handling rather than claiming 1,000 independent samples. A usefulness outcome could be correctly reproduced held-out defects or successful repairs; harmful suppression concerns

valid eligible work stopped by the policy. New domains need their own checked cases/near misses and renewed scope/evaluation evidence.

Q24

Accept: "Three educational checks passed; eleven source-dependent checks were skipped. The PDF compiled and the declared file hashes matched. These results do not execute the production behavior, establish detector effectiveness, or create a Zenodo publication." State the skip cause, needed source checkout and test executions, exact hashes, and publication status. Hash identity, exercised behavior, effectiveness, and external publication are separate claims. An honest methods preprint can state those limits.

Replay and rubric

Run the standard-library exact worksheet:

```
python code/landau_worksheet.py
```

It recalculates the quiz's fractions, batching endpoints, cost count, normalization example, key-order counterexample, and catalog counts. It does not replay the production sink, certify JCS, or run a detector. The repository-dependent storage tests supply the separately scoped behavior checks. Follow the study guide for both kinds of replay.

Award one point each for result, derivation/counterexample, declared contract, and evidence limit. Require at least 3/4 on every question in each completed level, with the quiz's evidence-boundary corrections before advancement. Alternative valid packets or experiments receive credit when they cover the same obligations without inventing observed results.